Full Comparison Report: Sensor A Channel 3 (g) vs Sensor B accZ(g)

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Time Domain Statistics

Metric	Sensor A	Sensor B	Difference
mean	-0.000006	-0.000027	0.000020
std	0.036617	0.051602	-0.014985
rms	0.036617	0.051602	-0.014985
peak	0.182414	0.402612	-0.220197
peak_to_peak	0.362620	0.733550	-0.370930
crest_factor	4.981638	7.802232	-2.820594

Correlation Analysis

- Pearson r: 0.007457
- R^2: 0.000056
- Spearman r: -0.003441

Error Metrics

- Pearson r: 0.007457 [95% CI: -0.002972, 0.016835]
- MAE: 0.048712 [95% CI: 0.048364, 0.049059]
- RMSE: 0.063051 [95% CI: 0.062604, 0.063543]
- NRMSE: 0.173876
- Max Error: 0.388504

Coherence Analysis

- Mean coherence: 0.0481 (Poor)
- Min coherence: 0.0000

Bland-Altman Analysis

- Mean difference: 0.000020

- Std difference (ddof=1): 0.063052
- Upper LoA: 0.123599
- Lower LoA: -0.123558

Normality Test (Shapiro-Wilk)

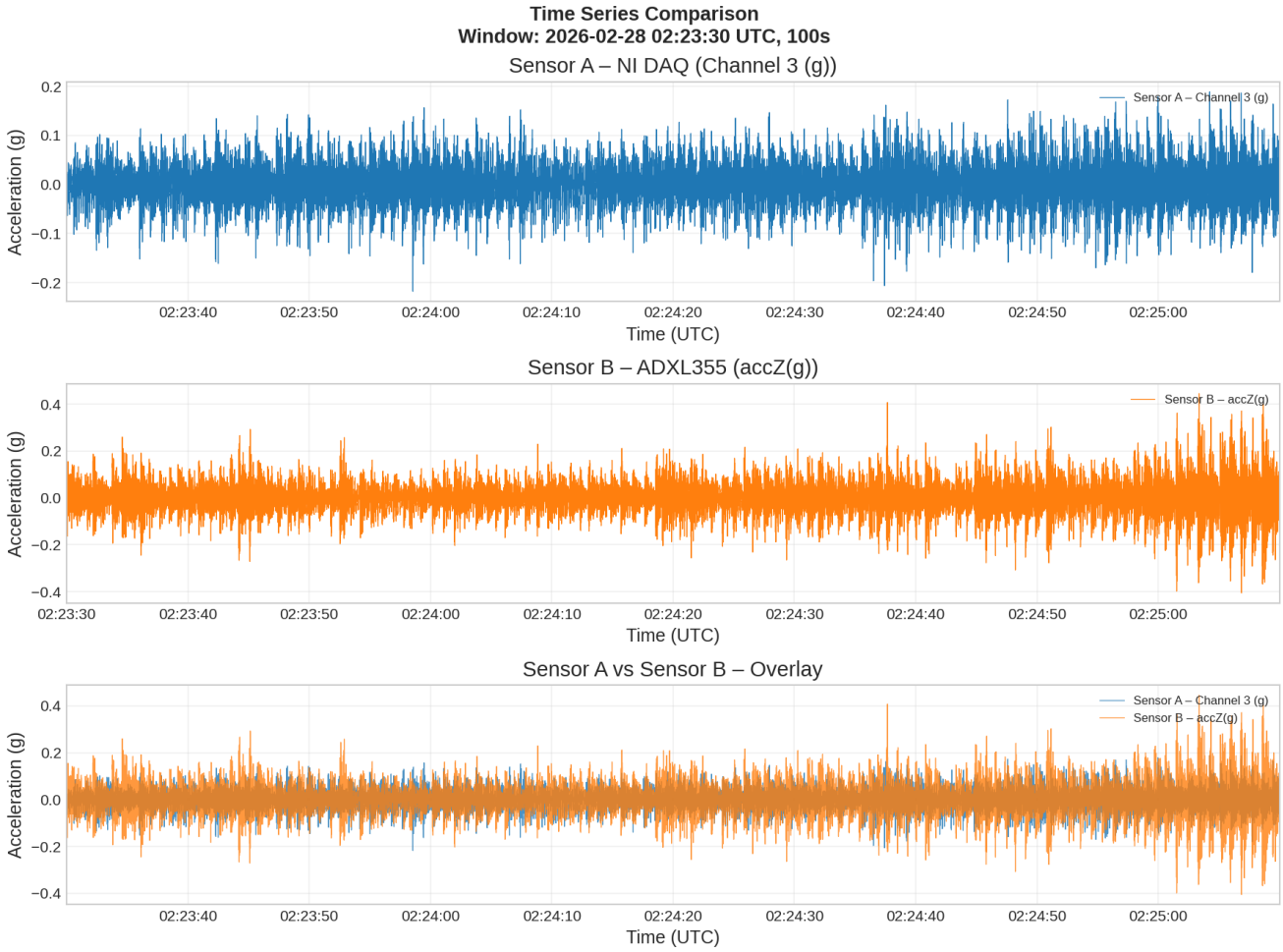
- Statistic: 0.995030
- p-value: 0.000000
- Result: Differences are **not normally distributed** (p <= 0.05) — LoA should be interpreted with caution

Frequency Domain Analysis

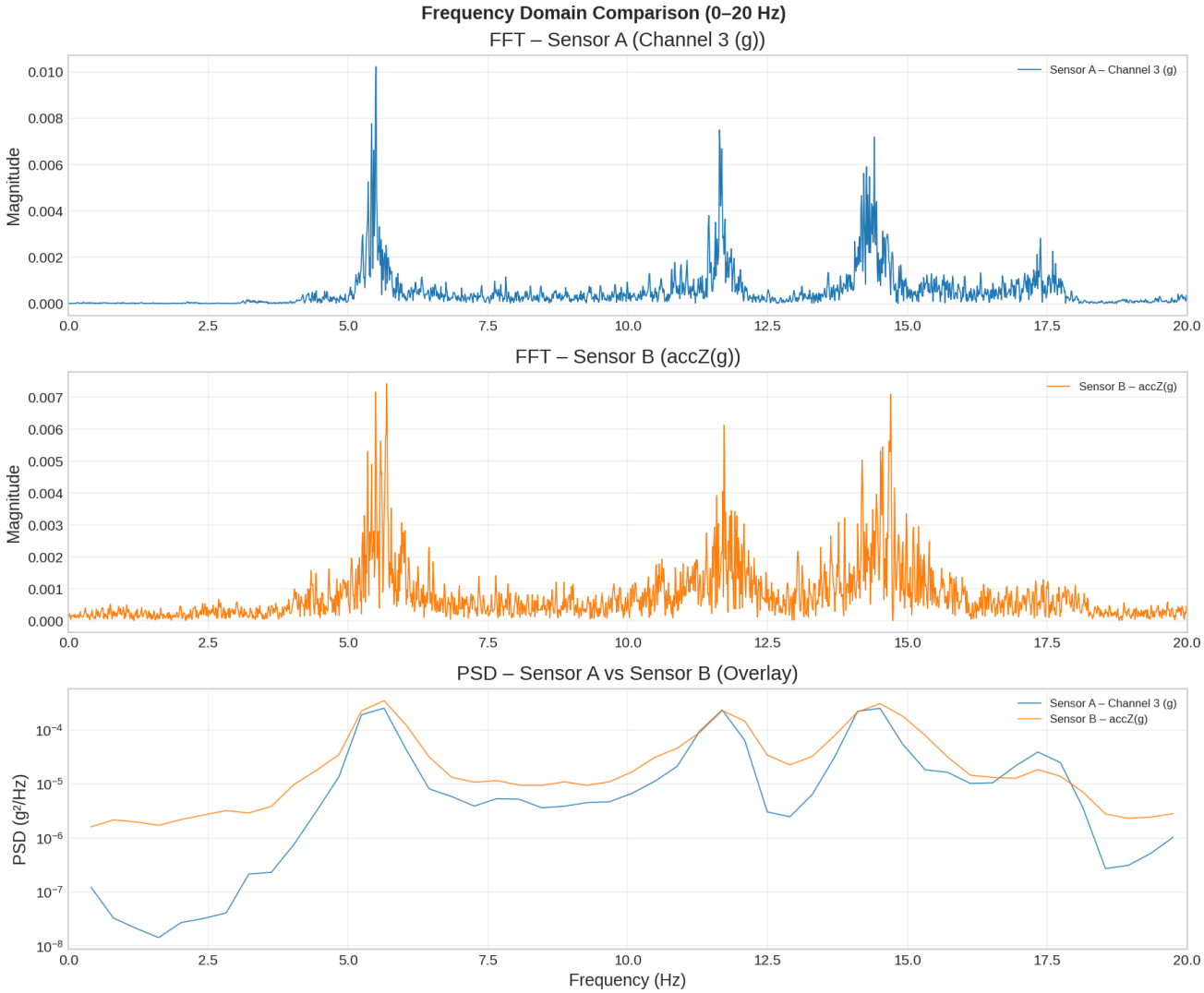
Metric	Sensor A	Sensor B
dominant_frequency	5.50 Hz	5.69 Hz
spectral_centroid	36.45 Hz	62.73 Hz
median_frequency	15.72 Hz	42.35 Hz

Figures

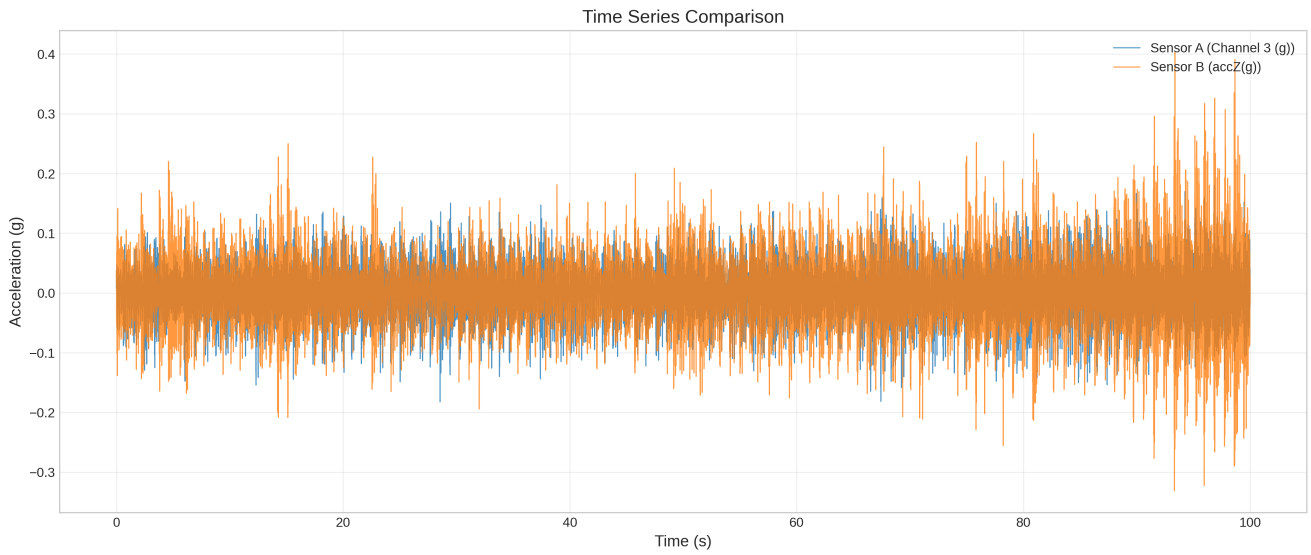
Time Series — Sensor A, Sensor B, Overlay (UTC axis)



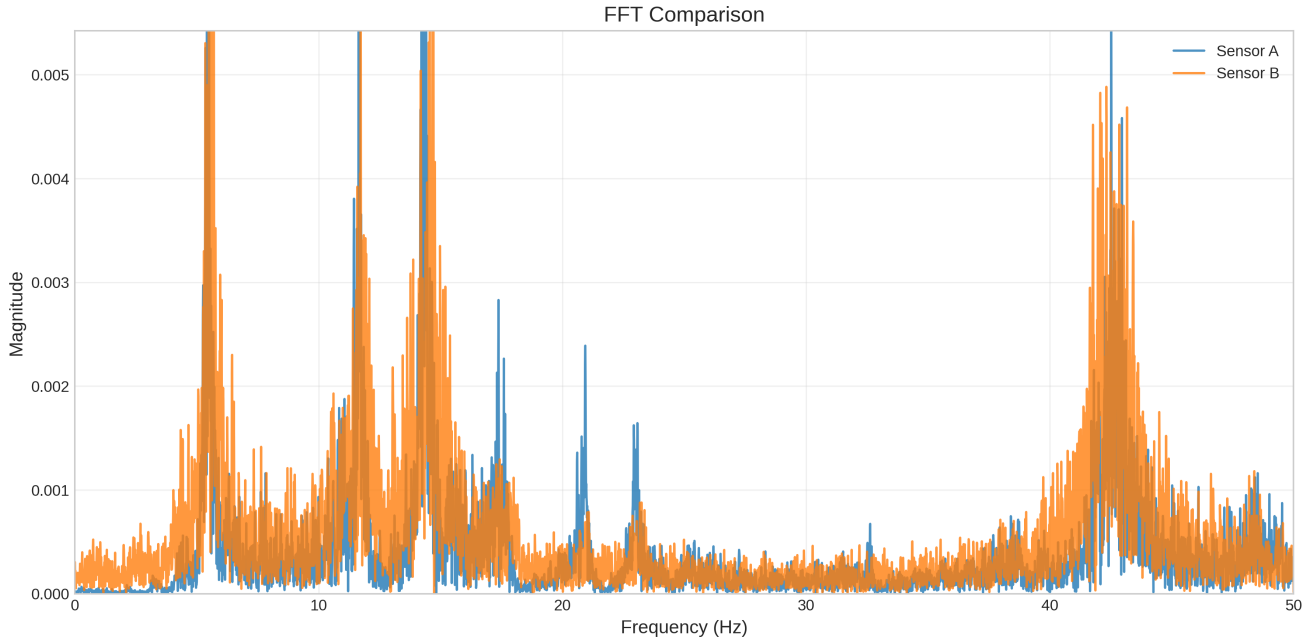
Frequency Domain — FFT A, FFT B, PSD Overlay (0–20 Hz)



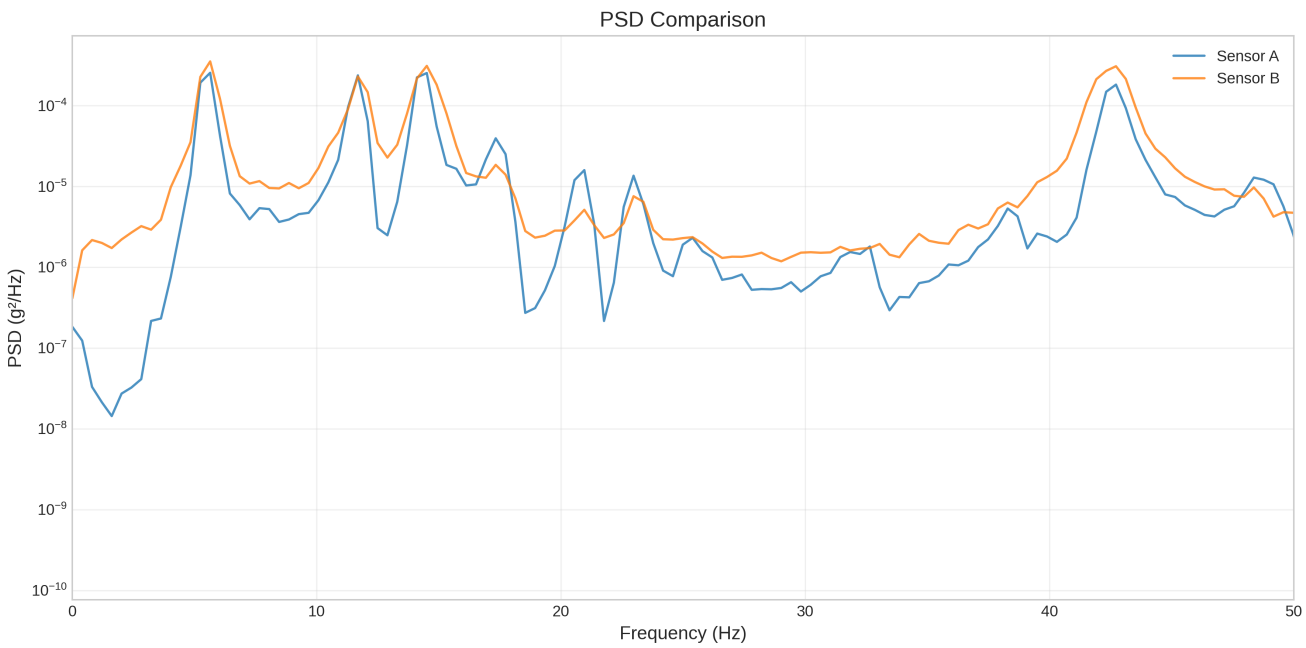
Time Series — Overlay (resampled)



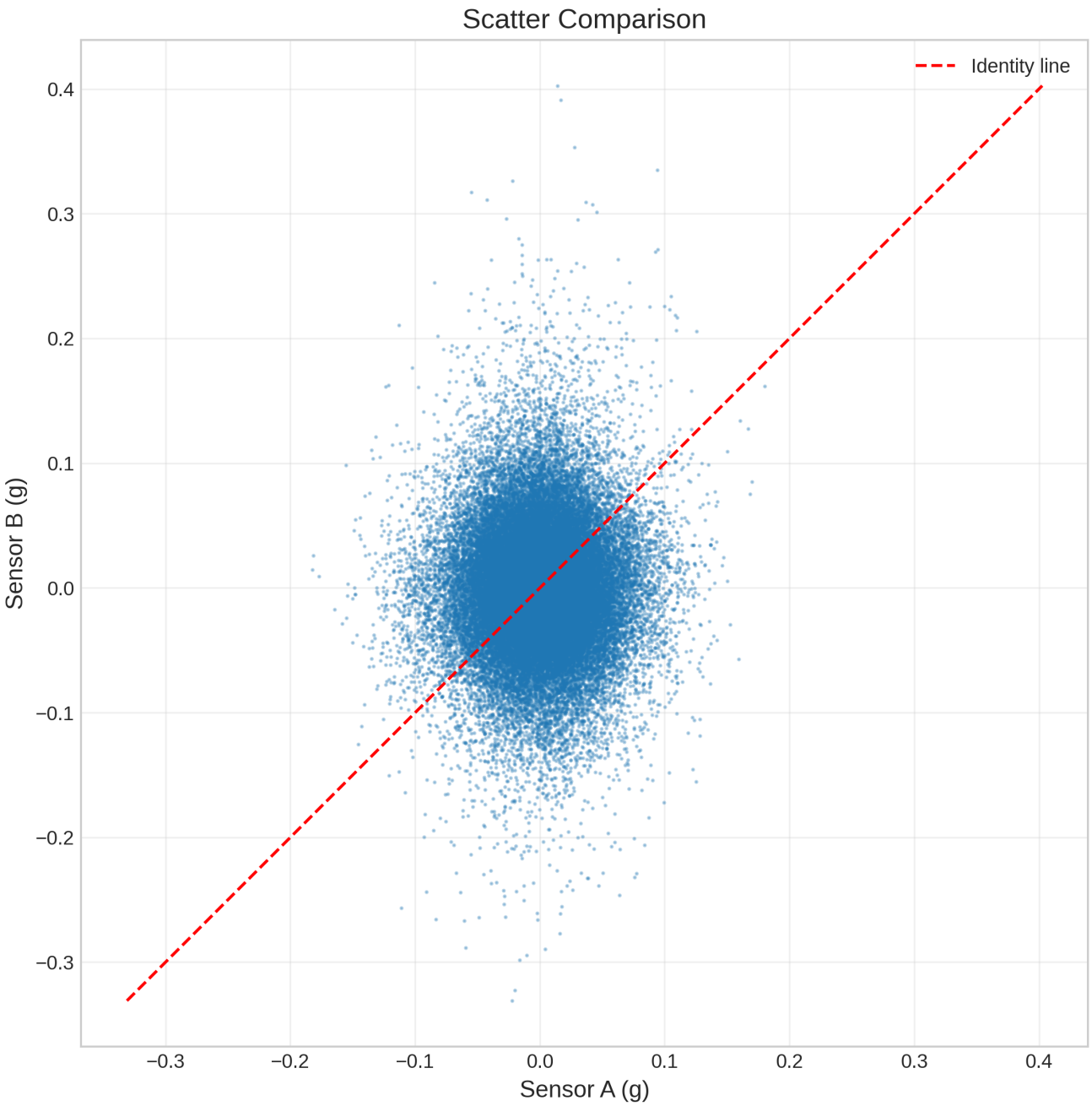
FFT Comparison



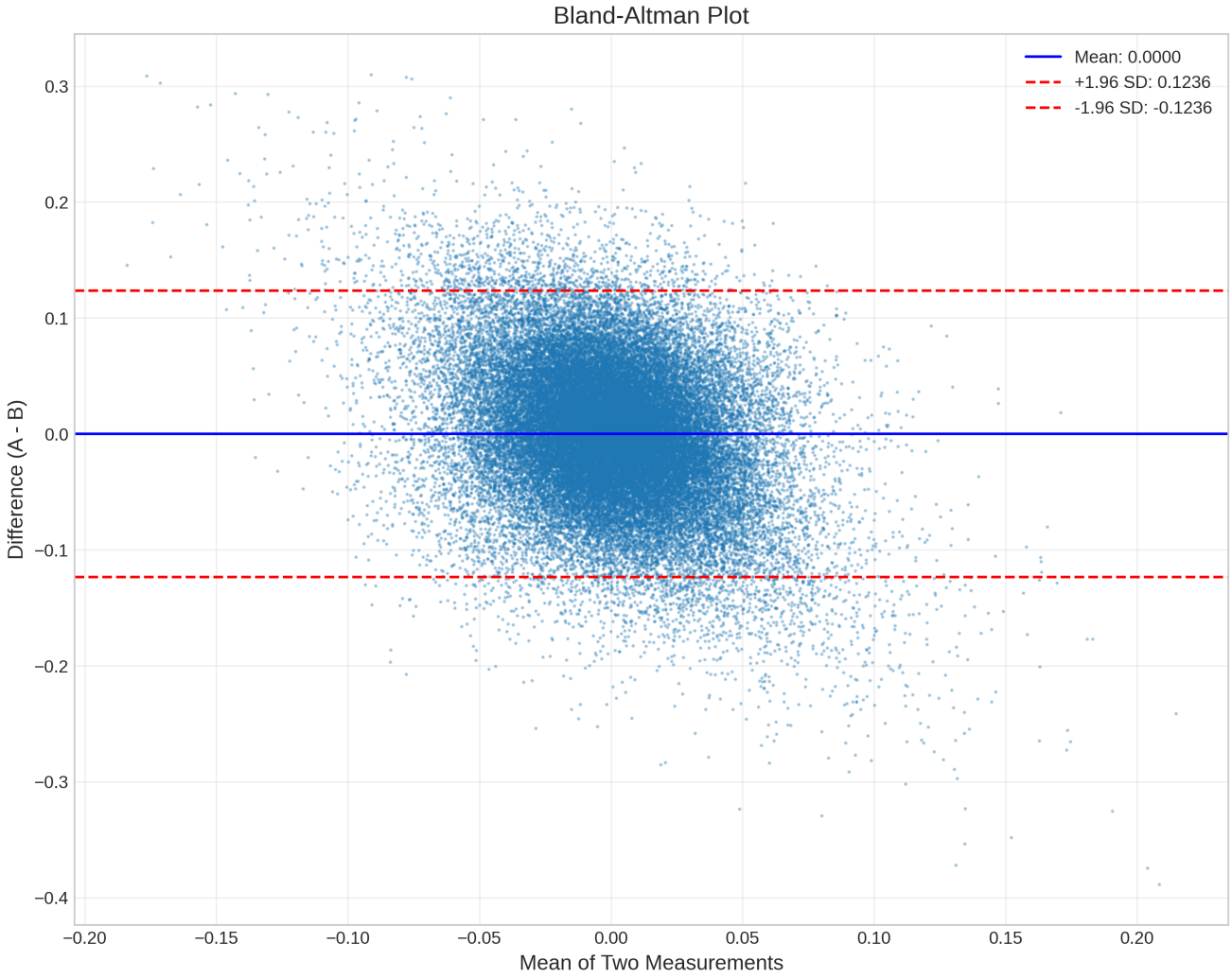
PSD Comparison



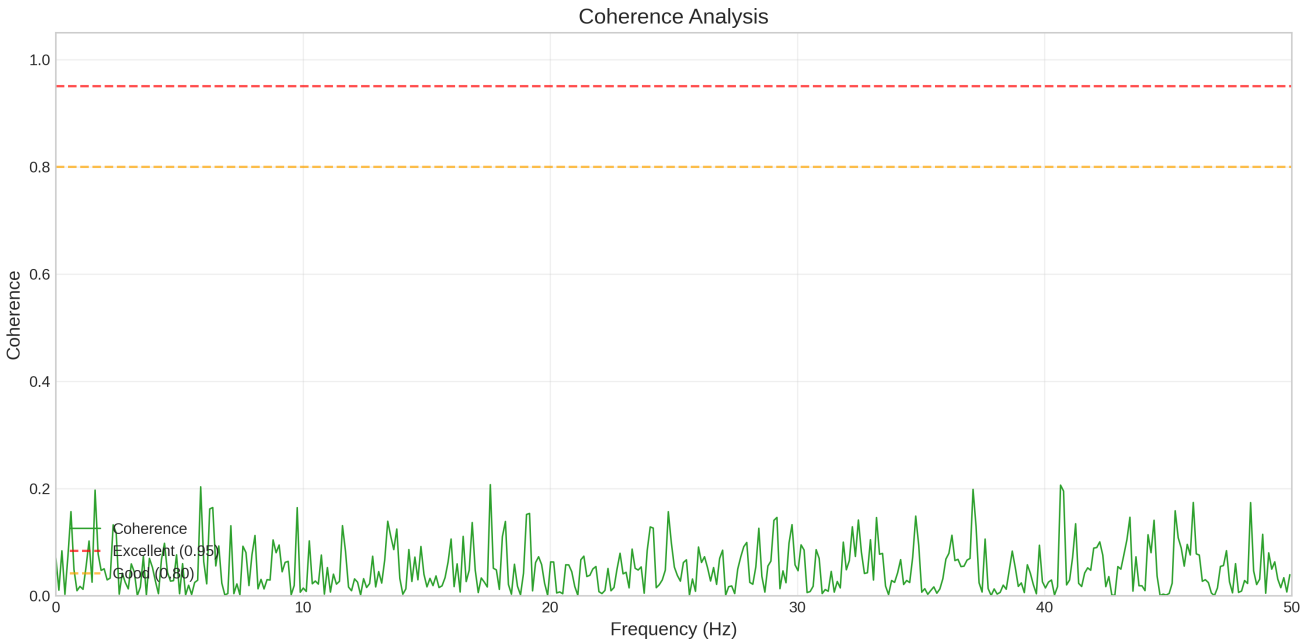
Scatter Plot



Bland-Altman Plot



Coherence Analysis



Frequency Band Analysis (0–20 Hz)

Per-sensor Statistics

Metric	Sensor A	Sensor B	Difference
mean (magnitude)	0.000525	0.000778	-0.000252
std (magnitude)	0.000863	0.000866	-0.000004
rms (magnitude)	0.001010	0.001164	-0.000154
peak (magnitude)	0.010227	0.007425	0.002802
peak frequency (Hz)	5.500000	5.690000	-0.190000
spectral energy	0.002042	0.002710	-0.000668

- **Energy ratio (A/B)**: 0.7534

Spectral Comparison Metrics

Metric	Value
Pearson r (spectra)	0.564525
MAE (magnitude)	0.000508
RMSE (magnitude)	0.000845
NRMSE (magnitude)	0.082640

Interpretation

- **Correlation (Pearson r = 0.0075)**: Poor
- **NRMSE = 0.1739 (17.4%)**: Poor
- **Dominant frequency**: matched (5.50 Hz vs 5.69 Hz) — same vibration source confirmed

Conclusion: Both sensors capture the same dominant vibration frequency, but waveform correlation is low. This is consistent with sensors placed at different physical locations on the structure, where vibration amplitude and phase differ. Time-lag sweep (cross-correlation over full data) confirmed no single offset improves correlation significantly (peak $r < 0.08$ across all channels). The low correlation is therefore attributed to spatial differences in sensor placement, not to timestamp misalignment.